

Technical Test - AI/ML Engineer - Insignia

Context

We are looking for engineers who build systems that work in production, not tutorial showcases or documentation scaffolds. There is no single "correct" architecture here. What we evaluate is how you think, what decisions you make, and how you justify them.

Dataset

You will be provided with two PDF documents related to Indonesian labor law (both documents are included in the ZIP file provided alongside this test):

1. UU No. 13 Tahun 2003 tentang Ketenagakerjaan
2. PP No. 35 Tahun 2021 tentang PKWT, Alih Daya, Waktu Kerja, dan PHK

Important notes:

- Documents may be scanned PDFs or native PDFs, handle both accordingly
- Document pre-processing is part of this task, so you are responsible for the full pipeline from raw PDF to a functional system

Task

Build a Retrieval-Augmented Generation (RAG) system based on the two provided documents. The system must be able to answer user questions accurately in Bahasa Indonesia, with responses that are relevant and traceable to their source, at minimum, citing the document name and page number.

We do not prescribe architecture. We want to see a system that reflects **your understanding and experience in building RAG**, choose the approach you consider most appropriate, and explain your reasoning, and make sure it runs. Your system will be tested against a set of prepared questions as part of the evaluation.

What we look for:

- How well the system handles ambiguous or incomplete queries
- How you approach documents that are complex in structure
- Retrieval quality, whether what is retrieved is genuinely relevant
- How you handle the raw documents. from extraction to indexing
- How honestly the system communicates its limitations

Deliverables

1. GitHub Repository (Public)

Structure is your choice, but the system must be runnable with minimal setup instructions.

2. README

The README is a core part of the evaluation, not a formality. Address the following clearly:

- System architecture, what components exist and how they interact
- Design decisions, why you chose the approach you took
- Tradeoffs, what you gave up and why that was the right call
- Limitations, what does not work well or what you did not address
- What is next, given more time, what would you improve and why
- How to run, ensure your system is accessible via a clear interface (REST API, CLI, or simple UI) so it can be queried directly during evaluation

If you used an agentic coding tool (Claude Code, Codex CLI, Cursor, or similar) during development, describe how you set it up and structured your workflow, not just that you used it, but how you engineered the experience.

3. Architecture Diagram

Any format is acceptable (draw.io, Mermaid, hand-drawn), clarity and accuracy are what matter.

Submission

Send the repository link to muhammad.basil@insignia.co.id. Ensure the README is readable without needing to run the code first. Questions that genuinely block you from getting started are welcome. We will not, however, provide hints about the expected approach.

Good luck! We are genuinely looking forward to seeing what you build. 🚀